Distributed Training of Large Language Models: Architectures, Parallelism, and Communication Challenges

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References

IBentryALTinterwordspacing Z. Zhu, C. Giannoula, M. Andoorveedu, Q. Su, K. Mangalam, B. Zheng, and G. Pekhimenko, "Mist: Efficient distributed training of large language models via memory-parallelism co-optimization," in *Proceedings of the Twentieth European Conference on Computer Systems*, ser. EuroSys '25. ACM, Mar. 2025, pp. 1298–1316. [Online]. Available: http://dx.doi.org/10.1145/3689031.3717461